Mukesh Kumar Sahu

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EDUCATION

- **National Institute of Technology Silchar** Aug 2023 – June 2025
  M.Tech in Data Science and Engineering CGPA: 8.56
  Thesis: "Self-Constructing Graph Model for Predicting Patient Criticalness using GNNs on MIMIC-III Data"
- **Silicon Institute of Technology** Aug 2016 – Jul 2020
  B.Tech in Electrical and Electronics Engineering CGPA: 7.82

WORK EXPERIENCE

- **AI/ML Intern, Tata Electronics, Bengaluru, India** Mar 2025 – June 2025
  - Architected and deployed LSTM Autoencoder pipeline for multivariate Statistical Process Control (SPC) parameter analysis, achieving 95% anomaly detection accuracy and reducing false positives by 40%
  - Implemented hybrid LSTM-Transformer architecture for predictive maintenance, enabling 10-minute advance CpK (Process Capability Index) drop prediction with 92% precision
  - Engineered real-time MLOps dashboard using Python and Dash, monitoring SPC and Cp/Cpk metrics for 5+ production lines with automated alerting for anomalies
- **System Engineer, Tata Consultancy Services, Bengaluru, India** Mar 2021 – Jun 2023
  - Orchestrated 15+ Apache Airflow DAGs for automated data workflows, reducing processing time by 80%
  - Spearheaded data engineering migration from legacy FORTRAN to Python-based data pipelines, improving code maintainability and enabling 3x faster feature deployment
  - Architected AWS S3 Data Lake solution managing 100TB+ datasets, partitioning and optimizing query patterns to reduce data retrieval costs by 45%
  - Designed and implemented Spark-based ETL pipelines in Databricks, processing 10M+ records daily with 99.9% data quality assurance
  - Collaborated with UAT teams to deliver actionable data insights, resulting in a 25% improvement in decision-making accuracy

PROJECTS

- **GNN Framework for Healthcare AI - Patient Criticality Prediction (M.Tech Thesis)** Jul 2024 – May 2025
  - Pioneered a Similarity-Based Self-Constructing Graph Model (SBSCGM) using the MIMIC-III dataset (40,000+ patient records) to predict ICU patient criticality
  - Developed HybridGraphMedGNN achieving state-of-the-art performance: 94.2% AUC-ROC, 87.4% F1-score for mortality prediction
  - Implemented multi-hop reasoning, attention mechanisms and graph-based explainability modules, improving predictive performance by 15% over baseline GNNs
- **Used Car Price Prediction with Ensemble Learning** May 2024 – June 2024
  - Developed an ensemble learning model (XGBoost, LightGBM, Neural Networks) with hyperparameter tuning for used car price prediction, ranking in the top 18% (RMSE: 0.4898) in the KaggleX competition.
- **Computer Vision for Agricultural AI** Jan 2024 – Apr 2024
  - Built wheat disease classification system using transfer learning (MobileNet, EfficientNet, VGG19) achieving 98.8% accuracy on 10,000+ images with real-time inference at 15 FPS
  - Applied data augmentation and stratified sampling techniques to improve generalization by 12% and Integrated GradCAM visualizations for explainable AI to support user trust and model transparency

TECHNICAL SKILLS

Languages: Python, SQL **ML/DL:** Scikit-learn, PyTorch, TensorFlow, Keras, OpenCV, Statsmodels **Cloud/MLOps:** AWS (EC2, S3, SageMaker), Databricks, Spark, Airflow, MLflow, Docker, Kubernetes, Git, Jupyter **Databases:** PostgreSQL, MySQL, BigQuery **Deployment:** FastAPI, dash, shell scripting **Stats/Viz:** Regression, Classification, Clustering, Feature Engineering, Time Series, Pandas, NumPy, Matplotlib, Seaborn, Power BI

CERTIFICATIONS

- **Generative AI and LLMs: Architecture and Data Preparation** 01/2025 – Present
- **IBM Data Science Professional Certificate** 04/2024 – Present
- **AWS Cloud Quest: Cloud Practitioner** 07/2023 – Present

ACHIEVEMENTS

- **Xcelerate Warrior Certificate (12/2022)** • **Star of the Month Award (twice) (11/2021, 11/2022)**
- **DCA WINGS 1 (12/2021)**